

Body Control Module (BCM)

AUTOSAR High-Level Design Document

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1. Introduction

This document describes the High-Level Design (HLD) of the Body Control Module (BCM) software, developed in compliance with the AUTOSAR Classic Platform architecture. The BCM is responsible for managing body-domain functions including central locking, lighting control, window lift, and wiper control.

1.1 Scope

This HLD covers the Software Component (SWC) architecture, port interfaces, signal definitions, runnable entities, and inter-component data flows for the BCM ECU. It does not cover low-level driver implementation details, which are documented separately in the MCAL Integration Guide.

1.2 Referenced Documents

Doc ID	Title	Version
SRS-BCM-001	Software Requirements Specification - BCM	2.1
ICD-BCM-VN-002	Interface Control Document - Vehicle Network	1.4
SWA-PLATFORM-010	Gen4 Platform Software Architecture	3.0

2. Software Component Architecture

The BCM software is decomposed into the following Application Software Components (SWCs), each responsible for a distinct body-domain function. Components communicate exclusively through AUTOSAR Sender-Receiver and Client-Server ports as defined in Section 3.

2.1 Component Overview

Component Name	Component ID	Type	Primary Responsibility
CentralLockingMgr	SWC-BCM-001	Application SWC	Manage lock/unlock requests and door state
LightingCtrl	SWC-BCM-002	Application SWC	Control exterior and interior lighting
WindowLiftMgr	SWC-BCM-003	Application SWC	Manage power window motor control and anti-pinch
WiperCtrl	SWC-BCM-004	Application SWC	Control wiper motor speed and intermittent modes
BCM_DiagMgr	SWC-BCM-005	Application SWC	Handle UDS diagnostic requests for BCM subsystems
VehicleStateMonitor	SWC-BCM-006	Application SWC	Aggregate vehicle state signals (speed, ignition)

2.2 Component Dependencies

CentralLockingMgr depends on VehicleStateMonitor for vehicle speed to enforce speed-based auto-lock. WindowLiftMgr depends on VehicleStateMonitor for ignition state to disable window operation when the ignition is off, unless the convenience-close feature is active. LightingCtrl depends on VehicleStateMonitor for ambient light sensor data routed through the vehicle state aggregation logic.

BCM_DiagMgr has read/write access to internal diagnostic data of all other components via dedicated Client-Server interfaces.

3. Port and Interface Definitions

3.1 CentralLockingMgr Ports

Port Name	Direction	Interface Type	Data Element / Operation
PPort_LockRequest	Provided	Sender-Receiver	LockRequestStatus (enum: LOCKED, UNLOCKED, PENDING)
RPort_DoorState	Required	Sender-Receiver	DoorState (bitfield, 4 doors)
RPort_VehicleSpeed	Required	Sender-Receiver	VehicleSpeed_kph (uint16)
PPort_DiagAccess	Provided	Client-Server	ReadLockStatus(), WriteLockOverride()

3.2 WindowLiftMgr Ports

Port Name	Direction	Interface Type	Data Element / Operation
PPort_WindowPos	Provided	Sender-Receiver	WindowPosition_pct (uint8, per door)
RPort_IgnitionState	Required	Sender-Receiver	IgnitionState (enum: OFF, ACC, RUN, START)
RPort_PinchDetect	Required	Sender-Receiver	PinchForce_N (uint16, from sensor SWC)
PPort_DiagAccess	Provided	Client-Server	ReadWindowPosition(), CalibrateWindow()

3.3 LightingCtrl Ports

Port Name	Direction	Interface Type	Data Element / Operation
PPort_LightState	Provided	Sender-Receiver	LightState (bitfield: head/tail/fog/hazard)
RPort_AmbientLight	Required	Sender-Receiver	AmbientLightLevel_lux (uint16)
RPort_IgnitionState	Required	Sender-Receiver	IgnitionState (enum)

4. Runnable Entities and Scheduling

Each SWC exposes one or more Runnable Entities mapped to OS tasks via the RTE. The table below summarizes runnable periodicity and trigger conditions.

Runnable Name	Owning SWC	Trigger	Period
Rte_CentralLock_10ms	CentralLockingMgr	Timed	10 ms
Rte_CentralLock_OnLockEvent	CentralLockingMgr	Data Received Event	Event-driven
Rte_WindowLift_20ms	WindowLiftMgr	Timed	20 ms
Rte_WindowLift_OnPinch	WindowLiftMgr	Data Received Event	Event-driven
Rte_Lighting_50ms	LightingCtrl	Timed	50 ms
Rte_Wiper_20ms	WiperCtrl	Timed	20 ms
Rte_Diag_OnRequest	BCM_DiagMgr	Client-Server Invocation	Event-driven
Rte_VehicleState_10ms	VehicleStateMonitor	Timed	10 ms

5. Functional Flow: Auto-Lock on Speed Threshold

The following describes the end-to-end functional flow for the vehicle speed-based auto-lock feature, a safety-relevant convenience function.

1. VehicleStateMonitor reads raw vehicle speed from the CAN gateway signal VSS_Speed and publishes VehicleSpeed_kph via RPort_VehicleSpeed at 10 ms intervals.
2. CentralLockingMgr's Rte_CentralLock_10ms runnable evaluates whether VehicleSpeed_kph exceeds the configured threshold (default: 15 kph, calibratable via BCM_DiagMgr).
3. If the threshold is exceeded and all doors report DoorState = CLOSED, CentralLockingMgr transitions internal state to PENDING and issues a lock command via PPort_LockRequest.
4. The lock actuator driver (outside SWC scope, handled by Complex Device Driver) executes the physical lock and reports completion, at which point LockRequestStatus transitions to LOCKED.
5. BCM_DiagMgr logs the auto-lock event with timestamp for UDS diagnostic readout (DID 0xF190 series).

6. Known Open Issues and Design Notes

ID	Description	Status
ISS-014	WindowLiftMgr does not currently subscribe to PinchForce_N from all four door sensor SWCs; only front doors are wired in current revision.	Open
ISS-021	LightingCtrl ambient light threshold for auto-headlamp activation is hardcoded; should be migrated to a calibratable parameter in next revision.	Open
ISS-027	No explicit timeout defined for PENDING lock state if actuator fails to confirm; potential gap for stuck-state diagnostic monitoring.	Open

7. Revision History

Version	Date	Author	Change Summary
1.0	2025-03-10	System Architecture Team	Initial release
2.0	2025-08-22	System Architecture Team	Added WiperCtrl and BCM_DiagMgr components
3.0	2026-02-15	System Architecture Team	Restructured ports for VehicleStateMonitor aggregation
3.2	2026-09-01	System Architecture Team	Added known issues section; updated auto-lock flow